

DIGITAL SYSTEM

A horizontal, glowing orange-yellow light streak that spans across the middle of the image, positioned below the title.

INTRODUCTION

- System = set of related components that work to achieve a goal



- Digital System = System in which signals have finite number of discrete values

- Digital systems are used to store, process and communicate information in digital form
 - Applications = process control system, communication system, digital instrument
 - Types =
 - Combinational Circuit
 - Sequential Circuit
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- Characteristics –

- Data is represented as a vector of **binary variables**
- Provide **accuracy in terms of number of bits** used to represent a variable
- Data representation is **suitable for error detection** and error correction
- Generates square waveform
- Digital signal is represented by two possible voltages on a wire – 0 volts i.e. low logic and 5 volts i.e. logic high

- Widespread use of Digital System –
 - Digital Circuit operates in one of two states – ON / OFF only
 - There are only few basic operations which are easy to understand
 - Requires Boolean Algebra which is very easy to learn
 - Large number of ICs are available to perform various operations

NUMBER SYSTEM

- System of writing to express numbers
- Types –
 - Decimal Number System
 - Binary Number System
 - Octal Number System
 - Hexadecimal Number System
- Signed Binary Number –
 - Sign Magnitude Representation
 - One's Complement Representation
 - Two's Complement Representation

- Binary Arithmetic –
 - Binary Addition
 - Binary Subtraction
 - Binary Multiplication
 - Binary Division
 - 2's Complement Arithmetic –
 - Addition / Subtraction using 2's Complement
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CODE

- Various binary codes used to represent data (numeric, alphabets or special characters)
- Examples –
 - Numeric Code –
 - Straight Binary Code
 - Natural BCD
 - Excess-3 Code
 - Gray Code
 - Alphanumeric Code –
 - Extended BCD Interchange Code
 - American Standard Code for Information Interchange

BASIC DIGITAL CIRCUITS

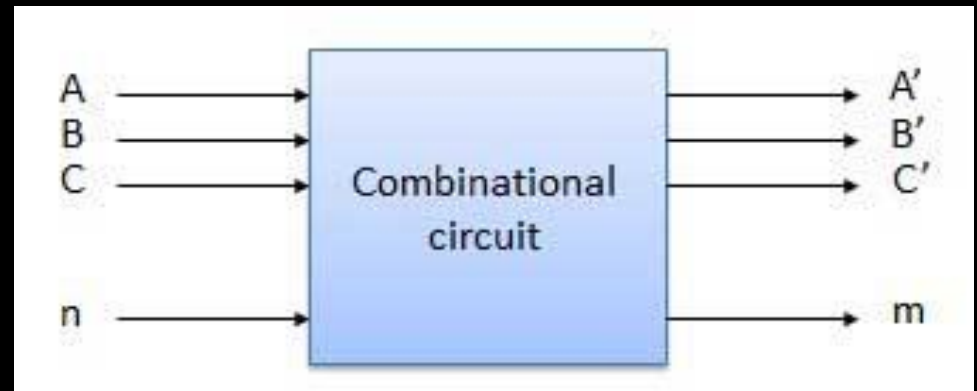
- Logic Gates – building blocks of any digital system
 - AND
 - OR
 - NOT
 - EX-OR
 - NAND
 - NOR
 - EX-NOR

BOOLEAN ALGEBRA

- Boolean Algebra used to analyze and simplify the digital circuits
- Boolean Laws –
 - Commutative Law
 - Associative Law
 - Distributive Law
 - Idempotent Law
 - Complement Law
 - Absorption Law
 - De'Morgan's Law

COMBINATIONAL LOGIC CIRCUIT

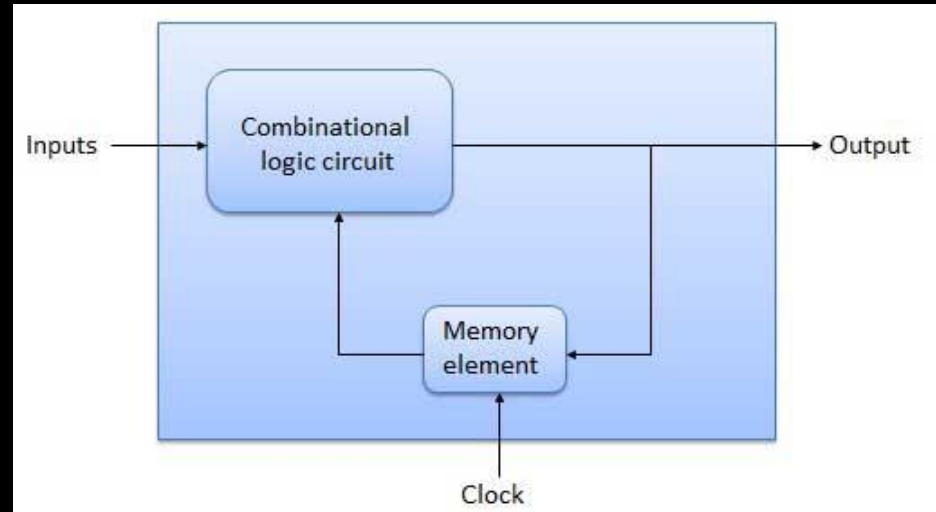
- Contain different types of Logic Gates
- Output is determined from present combination of inputs



- Example –
 - Adder / Subtractor
 - Decoder / Encoder
 - Multiplexer / De-Multiplexer

SEQUENTIAL LOGIC CIRCUIT

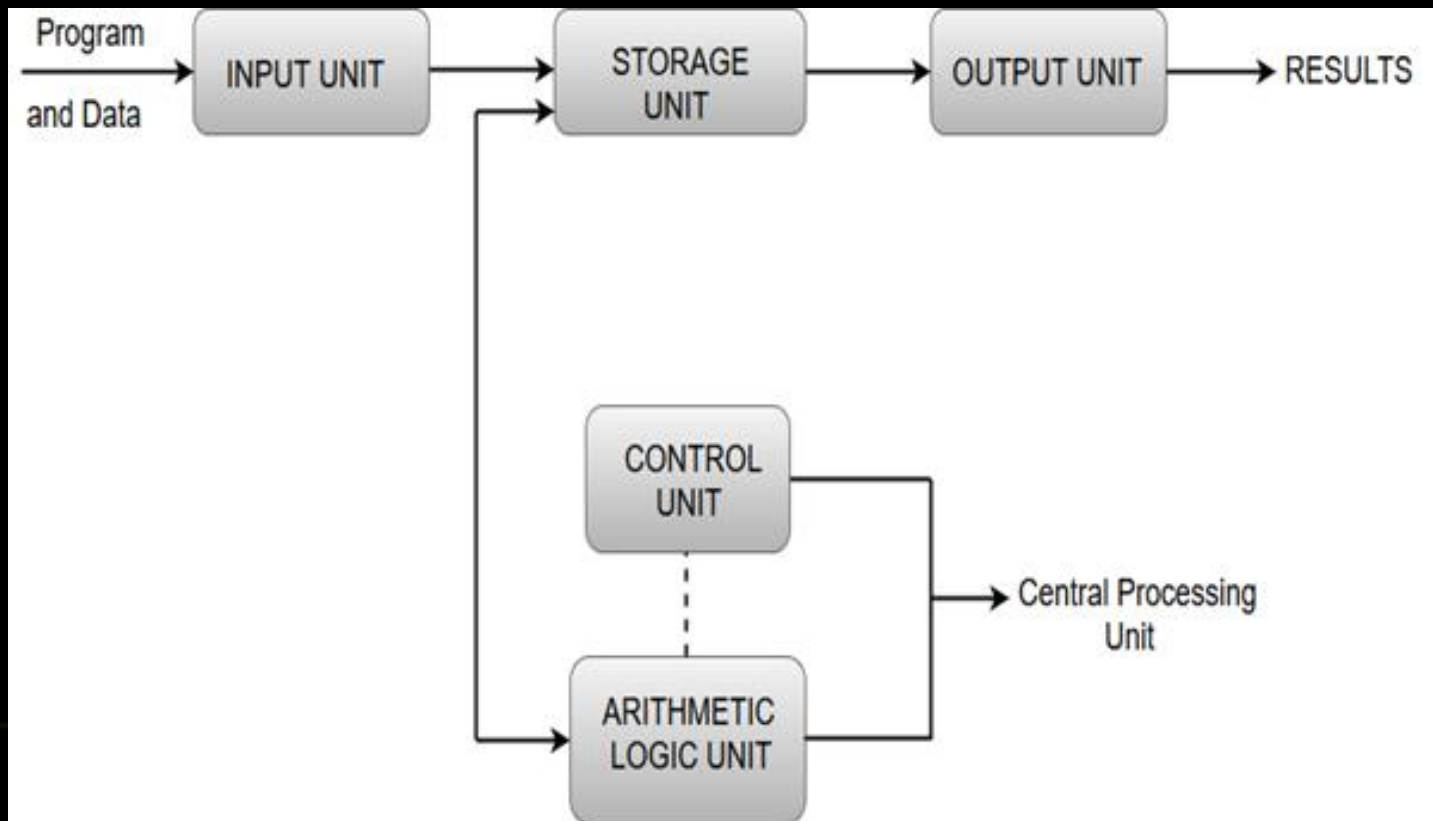
- Contain different types of Logic Gates
- Output is determined from present combination of inputs as well as previous output



- Example –
 - Flip-flop
 - Shift Register
 - Digital Counters

FUNCTIONAL UNITS OF DIGITAL SYSTEM

- A computer organization describes functions and design of various units of digital system



- Von Neumann Architecture
- Basic Instruction Cycle
- Memory Hierarchy

MICROPROCESSOR

- IC – capable of performing arithmetic and logic operations
- 8086 microprocessor –
 - Functional Block Diagram
 - Instruction Format
 - Addressing Modes
 - Segmented Memory and Interleaved Memory